

FIT COMPETITION 2026

Track II: *Infrastructure and Energy/Power Failure*

ANOMALY DETECTION IN PEER-TO-PEER SMART GRID BLOCKCHAIN TRANSACTIONS USING UNSUPERVISED ENSEMBLE LEARNING



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Chapter I: Introduction

1.1 Background

Blockchain technology has emerged as a promising solution for enabling secure and transparent transactions in smart grid environments. By providing a decentralized and tamper-resistant ledger, blockchain allows participants to record and verify transactions without relying on a central authority [1]. In smart grids, blockchain serves as the trust layer for peer-to-peer (P2P) energy trading, where prosumers can directly exchange surplus energy with consumers while ensuring transaction integrity and traceability [2, 3]. This capability supports the development of more flexible and decentralized energy systems.

Despite its advantages, blockchain-based P2P energy trading generates large volumes of transactional data that are difficult to monitor manually. Abnormal transaction patterns may indicate suspicious activities or potentially fraudulent behavior that warrant further investigation [4]. Detecting such anomalies is challenging because suspicious patterns may exist within valid transactions, even when the blockchain itself remains uncompromised. Furthermore, the lack of anomaly labels in real-world blockchain datasets makes supervised learning difficult, motivating the use of unsupervised methods [5].

Previous studies have extensively investigated blockchain applications in smart grids [1, 2, 3] as well as anomaly detection in blockchain systems [4, 5]. However, while anomaly detection has been widely studied in blockchain systems, its application to blockchain-based P2P energy trading environments remains relatively underexplored. Existing studies primarily focus on transactional patterns, with limited investigation of combined transactional, network, and blockchain structural characteristics.

To address this gap, this study proposes an ensemble anomaly detection framework that combines Isolation Forest, Local Outlier Factor (LOF), and One-Class Support Vector Machine (One-Class SVM). By combining multiple unsupervised detection techniques, the proposed approach aims to improve the robustness of anomaly identification in blockchain-enabled smart grid transactions.

1.2 Problem Statement

Based on the background described above, the problems addressed in this study are:

1. Blockchain-based P2P smart grid environments generate large volumes of transactional data without anomaly labels, making it difficult to identify sus-

picious patterns using conventional supervised approaches.

2. It remains unclear which transactional, network-based, and blockchain structural characteristics are most strongly associated with anomalous behavior in P2P energy trading systems.
3. The effectiveness of combining multiple unsupervised detectors through an ensemble voting strategy for improving anomaly detection robustness in this context has not been thoroughly demonstrated.

1.3 Objectives

The objectives of this study are:

1. To develop an unsupervised ensemble anomaly detection framework combining Isolation Forest, Local Outlier Factor, and One-Class SVM for identifying anomalous transactions in blockchain-based P2P smart grid data.
2. To analyze and identify transactional, network-based, and blockchain structural features that contribute most significantly to distinguishing anomalous from normal trading behavior.
3. To evaluate the detection robustness of the majority-vote ensemble strategy relative to each individual model through score distribution analysis, cross-model agreement, and feature discriminative power assessment.

1.4 Benefits

The expected benefits of this study are as follows:

1. To provide a practical framework for anomaly detection in blockchain-based smart grid systems where labeled anomaly data are unavailable.
2. To demonstrate the applicability of unsupervised machine learning techniques for monitoring and analyzing P2P energy trading transactions.
3. To contribute empirical insights that support future research on blockchain-enabled smart grid analytics and anomaly detection.

Chapter II: Literature Review

2.1 Data Science Concepts

2.1.1 Blockchain Technology in Smart Grids

Blockchain is a decentralized and immutable ledger that records transactions across distributed network participants through cryptographic mechanisms [1]. Each block contains validated transactions and a cryptographic reference to the previous block, ensuring transparency, traceability, and resistance to unauthorized modification. These characteristics make blockchain particularly suitable for environments requiring secure and auditable transaction records.

In smart grid environments, blockchain serves as the trust infrastructure that enables peer-to-peer (P2P) energy trading among distributed participants [2]. Smart contracts further automate transaction verification and settlement processes, reducing dependence on centralized intermediaries while improving operational efficiency [3]. As a result, blockchain has become a key enabling technology for decentralized energy systems and modern smart grid applications.

2.1.2 P2P Energy Trading Networks

Peer-to-peer (P2P) energy trading enables prosumers, entities capable of both generating and consuming electricity, to directly exchange surplus energy with consumers through decentralized market mechanisms [2]. The increasing adoption of distributed renewable energy resources, particularly solar photovoltaic systems, has accelerated interest in P2P trading as an alternative to conventional centralized electricity markets.

P2P energy trading offers several benefits, including improved utilization of renewable energy resources, reduced transmission losses, and greater flexibility in energy distribution. However, the involvement of numerous participants also introduces challenges related to transaction integrity, trust, and security. Blockchain technology addresses these concerns by maintaining a transparent and tamper-evident record of all trading activities within the network [1].

2.1.3 Anomaly Detection in Transactional Data

Anomaly detection is the process of identifying observations that deviate significantly from expected patterns or normal behavior. In transactional systems, anomalies may indicate fraudulent activities, abnormal participant behavior, operational faults, or data inconsistencies [5]. Detecting such irregularities is particularly important in blockchain-based smart grid environments because abnormal transactions may affect both financial exchanges and energy distribution processes.

Unlike traditional cybersecurity problems that focus on direct system compromise, anomalies in blockchain environments often occur within otherwise valid transactions [4]. Examples include unusual transaction values, unusual trading frequencies, or unexpected interaction patterns among participants. Furthermore, the scarcity of labeled anomaly data in real-world blockchain datasets [5] makes unsupervised and semi-supervised detection approaches more practical than supervised classification methods.

2.2 Machine Learning Methods

2.2.1 Isolation Forest

Isolation Forest isolates observations through recursive random partitioning of the feature space. Since anomalous points are typically rare and different from normal observations, they require fewer partitions to be isolated and therefore receive higher anomaly scores [6]. The method is computationally efficient and suitable for high-dimensional transaction datasets.

2.2.2 One-Class Support Vector Machine

One-Class SVM (OC-SVM) learns a decision boundary that encloses normal observations in a transformed feature space [7]. Transactions located outside this boundary are classified as anomalies.

2.2.3 Local Outlier Factor

Local Outlier Factor (LOF) detects anomalies by comparing the local density of a transaction to that of its nearest neighbors [8]. Transactions located in substantially lower-density regions receive higher anomaly scores. This enables LOF to identify local anomalies that may not appear abnormal from a global perspective.

2.2.4 Ensemble-Based Anomaly Detection

Ensemble anomaly detection combines multiple detectors to improve robustness and reduce model-specific weaknesses [5]. Voting-based ensemble strategies are commonly used to aggregate predictions from multiple detectors and improve detection reliability.

2.3 Related Works

2.3.1 Blockchain Applications and Frameworks for Smart Grids

Previous studies have demonstrated the potential of blockchain for enabling secure and decentralized smart grid operations. Surveys by Musleh et al. [1] and Andoni et al. [2] identified applications including P2P energy trading, distributed energy management, and grid resilience. These works emphasize blockchain's role in establishing trust and transparency but do not focus on anomaly detection within transaction streams.

2.3.2 Machine Learning for Anomaly Detection in Blockchain Systems

Machine learning has become an increasingly important tool for identifying suspicious behavior in blockchain transaction data. Farrugia et al. [4] demonstrated that data-driven approaches can effectively detect anomalous transaction patterns that are difficult to identify through manually defined rules. Their findings emphasize the importance of behavioral analysis in addition to blockchain's structural security mechanisms.

Patel et al. [5] further highlighted challenges including limited labeled datasets, evolving transaction behavior, and the need for scalable detection approaches. The review identified unsupervised and ensemble-based methods as promising solutions for practical blockchain anomaly detection scenarios.

2.3.3 Integration of Blockchain and AI for Network Security

Recent research has explored the integration of blockchain technology and artificial intelligence to strengthen security in intelligent network environments [10]. While blockchain ensures data integrity, traceability, and transparency, AI techniques provide the capability to analyze behavioral patterns and identify potentially malicious activities.

This combination is particularly relevant in smart grid systems, where secure transaction recording alone is insufficient to identify abnormal operational behavior. AI-based anomaly detection can complement blockchain infrastructure by providing an additional layer of monitoring and decision support. The present study contributes to this direction by applying an unsupervised ensemble anomaly detection framework to blockchain-based P2P energy trading transactions.

Chapter III: Methodology

3.1 Dataset Overview

The dataset simulates transactions on a Peer-to-Peer (P2P) energy trading network running on a blockchain, where producers and consumers exchange energy directly and every action is recorded as a transaction inside a block. Each block links to the previous one through a cryptographic hash, forming a chain of records. This structure means that, in addition to the transaction-level details, the dataset also carries block-level metadata that reflects how the blockchain itself was constructed during the simulation.

Two transaction types appear in the data: `energy_delivery`, where the grid operator sends energy to a consumer, and `energy_payment`, where a consumer pays a producer for that energy. Together, these two transaction types form a complete trading cycle: energy physically moves from the grid operator to a consumer, while the corresponding payment moves from the consumer to the producer who supplied that energy. Since no labeled anomalies exist, this project treats the task as unsupervised anomaly detection, aiming to identify transactions that deviate from normal trading behavior without relying on predefined fraud or fault definitions.

3.2 Data Description

- Number of records: 10,206 rows, including 1 genesis block row and 10,205 transaction rows.
- Features / variables: 13 columns, including `block_index`, `block_timestamp`, `block_miner_id`, `tx_id`, `tx_timestamp`, `tx_type`, `tx_sender`, `tx_recipient`, `tx_amount`, `tx_energy`, `block_hash`, `block_previous_hash`, and `block_nonce`.
- Data types: numeric (`tx_amount`, `tx_energy`, `block_nonce`, timestamps) and categorical/string (`tx_type`, `tx_sender`, `tx_recipient`, `block_hash`, `block_miner_id`).

The `tx_type` column contains 5,193 `energy_delivery` and 5,012 `energy_payment` transactions, an almost even split that reflects the paired nature of deliveries and payments in the network. `tx_amount` ranges from 0.002 to 1.015 (mean 0.178), and `tx_energy` ranges from 0 to 4.883 (mean 0.445), with zero values for payment transactions that carry no energy value. The wide range of `tx_energy` compared to `tx_amount` suggested early on that the two variables would need to be examined separately rather than assumed to scale together.

Table 3.1 shows the first eight rows. The `block_hash`, `block_previous_hash`, and `raw_timestamp` columns are omitted for readability, and `tx_id` values are truncated to eight characters. The sample illustrates the paired structure described above: block 1 contains one `energy_delivery` transaction from GRID-OP-01 to CON-12, immediately followed by the corresponding `energy_payment` from CON-12 to PRO-05 with the same amount, while block 2 contains multiple such pairs grouped together.

block_index	block_miner_id	tx_id	tx_type	tx_sender	tx_recipient	tx_amount	tx_energy	block_nonce
0	genesis	—	—	—	—	—	—	886
1	GRID-OP-01	21eed025...	energy_delivery	GRID-OP-01	CON-12	0.1859	0.9164	292
1	GRID-OP-01	23268eea...	energy_payment	CON-12	PRO-05	0.1859	0.0000	292
2	GRID-OP-01	1b3cc40f...	energy_payment	CON-17	PRO-06	0.0214	0.0000	70
2	GRID-OP-01	1d990f95...	energy_delivery	GRID-OP-01	CON-01	0.1553	0.7457	70
2	GRID-OP-01	23376b2b...	energy_payment	CON-01	PRO-05	0.1553	0.0000	70
2	GRID-OP-01	24a3bda8...	energy_delivery	GRID-OP-01	CON-16	0.1171	0.5610	70
2	GRID-OP-01	28e22cf5...	energy_payment	CON-17	PRO-05	0.6614	0.0000	70

Table 3.1: Sample of the first eight rows of the dataset.

3.3 Methodology Pipeline

The overall workflow followed in this project consists of seven stages, starting from the raw dataset and ending with the evaluation of the final anomaly labels. Figure 3.1 summarizes this pipeline. The raw transaction data was first explored to understand its structure, then cleaned and scaled during preprocessing, after which a set of 17 features was engineered from the cleaned data. This set of features was passed in parallel to three independent anomaly detection models, whose outputs were combined through a majority-vote ensemble to produce the final anomaly labels, which were then evaluated using the metrics described in Section 3.7. The remaining sections of this chapter will describe this pipeline in detail.

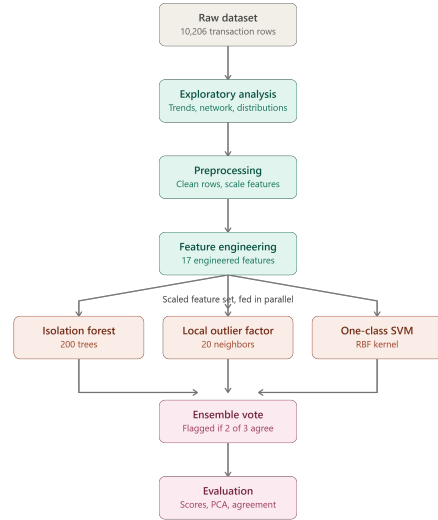


Figure 3.1: Overview of the methodology pipeline, from raw dataset to final evaluation.

3.4 Exploratory Data Analysis

Before modeling, the dataset was explored from three angles to guide feature engineering. This exploration step was important because the dataset does not come with documentation explaining what a "normal" transaction looks like, so these patterns had to be established directly from the data itself before any anomaly detection could be meaningful.

3.4.1 Time-Series Trends

Transactions were grouped into one-minute windows to examine transaction amount, energy delivered, and transaction count over time. This step helped reveal that transaction count follows a repeating cycle tied to block creation, rather than fluctuating randomly, and informed temporal features such as the hour and minute of each transaction and the elapsed time since the first transaction in the dataset.

3.4.2 Network Structure

Trading relationships were represented as a directed graph, with participants (producers, consumers, the grid operator) as nodes and transactions as edges. Building this graph made it possible to identify which participants act as central points in the network and supported graph-based features such as out-degree, in-degree, betweenness centrality, and PageRank, all of which describe how connected or central a participant is relative to the rest of the network.

3.4.3 Distribution and Role-Based Patterns

The distributions of `tx_amount` and `tx_energy`, and the relationship between the two, were examined, and transaction behavior was compared across roles (producer, consumer, grid operator) and transaction types. This clarified how `energy_delivery` and `energy_payment` transactions differ structurally, for instance, that only the grid operator sends `energy_delivery` transactions, and guided which features were worth constructing versus which would simply duplicate information already captured by `tx_type`.

3.5 Data Preprocessing

- Handling missing values: the genesis block row, which contained no transaction data, was removed, leaving 10,205 records for analysis.
- Encoding / normalization: numeric features were standardized using `StandardScaler` to zero mean and unit variance, ensuring that features measured on different scales, such as transaction amounts and graph centrality scores, contribute comparably to the distance and density calculations used by the anomaly detection models.
- Feature engineering: 17 features were built across four categories. Temporal

features include the hour and minute of each transaction and the elapsed time since the first transaction. Ratio and statistical features include the energy-to-amount ratio and z-scores of amount and energy. Rolling features include the per-sender rolling mean and standard deviation of amount, and the deviation of each transaction from its sender’s rolling mean. Network features include sender out-degree, recipient in-degree, sender betweenness centrality, and sender PageRank, all derived from the transaction graph described in Section 3.4.2. Block-level features include the log-transformed block nonce and the number of transactions per block.

The scaled features were also projected into two dimensions using PCA, retaining about 50.69% of total variance, for visualization only and not as a model input. While this projection does not capture the full structure of the 17-dimensional feature space, it provides a useful two-dimensional view for visually inspecting how anomalous and normal transactions are positioned relative to each other once the models have been trained.

3.6 Model / Method

Three unsupervised anomaly detection algorithms were used together and combined through a voting strategy, each with a contamination parameter of 0.015 (1.5%), representing the expected proportion of anomalies in the dataset. These three algorithms were deliberately chosen because they rely on different underlying principles, isolation, density, and boundary separation, so that an anomaly missed by one method has a reasonable chance of being captured by another.

3.6.1 Isolation Forest

Isolation Forest builds random decision trees and measures how easily each point can be isolated, since anomalies typically require fewer splits to separate from normal data. It was trained with 200 trees on the scaled features, producing an anomaly label and a continuous anomaly score for each transaction. This method serves as the primary global detector in the pipeline, as it considers all 17 features simultaneously when deciding how isolated a transaction is.

3.6.2 Local Outlier Factor

Local Outlier Factor (LOF) compares the local density around each point to its neighbors, flagging points in sparser regions as outliers. Configured with 20 neighbors in novelty mode, it captures local anomalies that Isolation Forest may miss, for example, a transaction that is only unusual relative to the typical behavior of a specific sender, rather than unusual across the dataset as a whole.

3.6.3 One-Class Support Vector Machine

One-Class SVM learns a boundary around the majority of the data using an RBF kernel, flagging points outside it as anomalies. It serves as a third, independent detector based on a different mathematical approach, providing a check on the results of the two other models from yet another perspective.

3.6.4 Ensemble Voting Strategy

The three models' outputs were combined by majority vote: a transaction was labeled anomalous only if at least two of the three models flagged it. This reduces false positives from any single model and reflects agreement across methods, so that the final anomaly label represents a consensus rather than the opinion of a single algorithm. The output of this stage is the final set of 85 ensemble-flagged transactions, which forms the basis for the results discussed in Chapter 4.

3.7 Model Evaluation

3.7.1 Evaluation Approach

Since no ground-truth labels exist, supervised metrics such as accuracy or F1-score could not be used. Evaluation instead relied on comparing anomaly score distributions, visualizing separation in PCA space, counting anomalies per model and by ensemble vote, measuring cross-model agreement, and comparing feature values between normal and anomalous transactions. Together, these approaches provide multiple independent lines of evidence for whether the flagged transactions represent a meaningful pattern rather than noise.

3.7.2 Metrics

- Anomaly score distributions: scores from Isolation Forest, LOF, and One-Class SVM compared between normal and ensemble-flagged transactions.
- Cross-model agreement: number of transactions flagged by each pair of models.
- Anomaly counts: number flagged by each model and by the final ensemble vote.
- Feature discriminative power: mean differences between normal and anomalous transactions per feature.

Chapter IV: Results and Discussion

4.1 Results

4.1.1 Exploratory Data Analysis Results

The time-series analysis in Figure 4.1 shows transaction count rising sharply and dropping close to zero in a repeating pattern. This reflects the blockchain's block creation cycle, where transactions accumulate before being grouped into a new block, rather than any irregularity in the data.



Figure 4.1: Time-series trends of mean transaction amount, mean energy delivered, and transaction count across one-minute bins.

The transaction network in Figure 4.2 consists of 31 nodes and 232 edges, with an average in-degree of 7.48. It follows a hub-and-spoke structure: the grid operator and consumers form a densely connected core, while the ten producers sit at the periphery, connected mainly back to this core. This makes the grid operator (GRID-OP-01) the central point through which most energy flows pass.

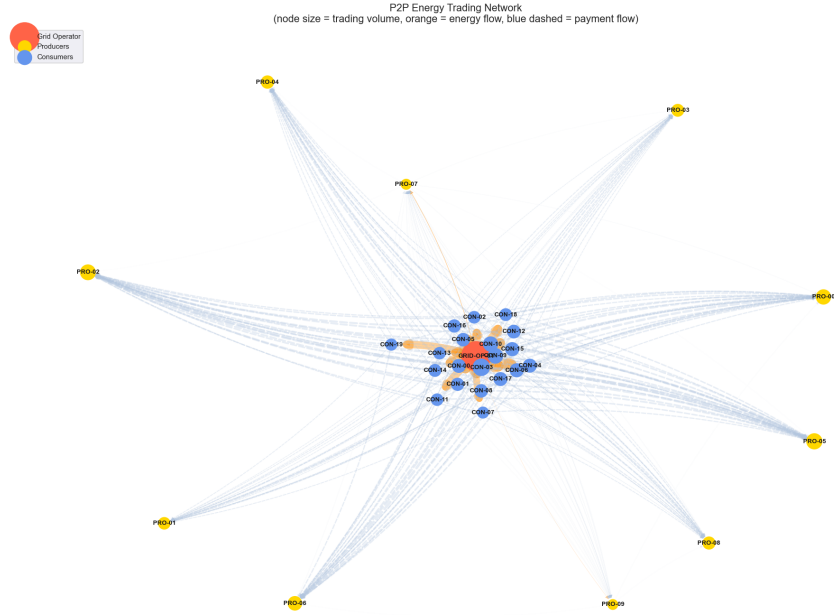


Figure 4.2: P2P energy trading network, with node size representing trading volume, orange edges representing energy delivery flow, and dashed blue edges representing payment flow.

Figure 4.3 shows the top ten sender-recipient pairs by total transaction amount, highlighting the highest-value trading relationships in the network.

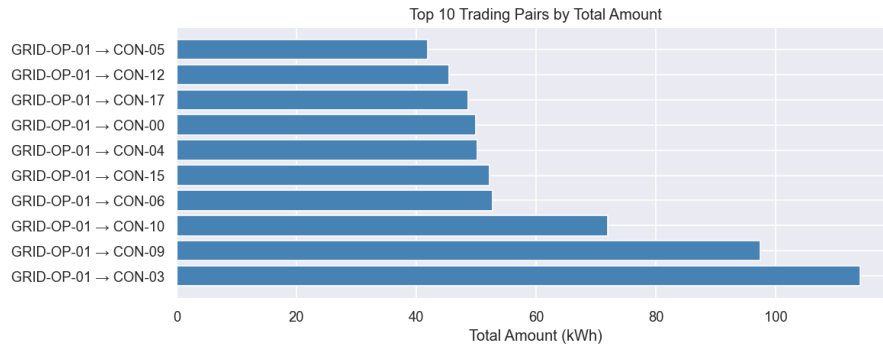


Figure 4.3: Top ten trading pairs ranked by total transaction amount.

The distributions of `tx_amount` and `tx_energy` in Figure 4.4 are both right-skewed, with most transactions involving small values and a smaller number forming a long tail of larger ones. Figures 4.5 and 4.6 show the amount-energy relationship by transaction type and behavior by participant role, confirming that energy_payment transactions always carry a `tx_energy` value of zero (financial settlement only), while energy_delivery transactions, which carry the actual energy value,

are sent only by the grid operator.

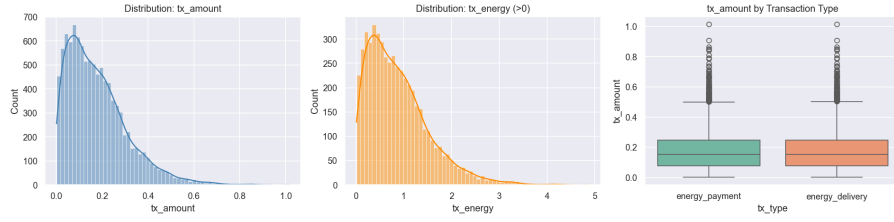


Figure 4.4: Distributions of transaction amount and transaction energy, and transaction amount grouped by transaction type.

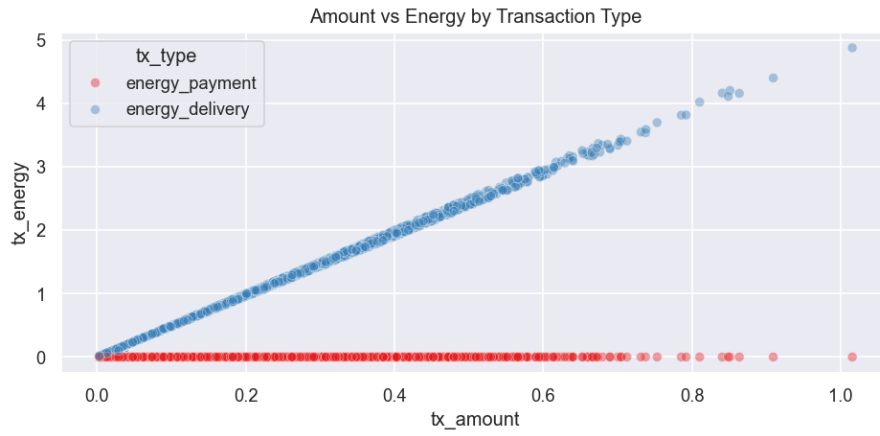


Figure 4.5: Relationship between transaction amount and transaction energy, separated by transaction type.

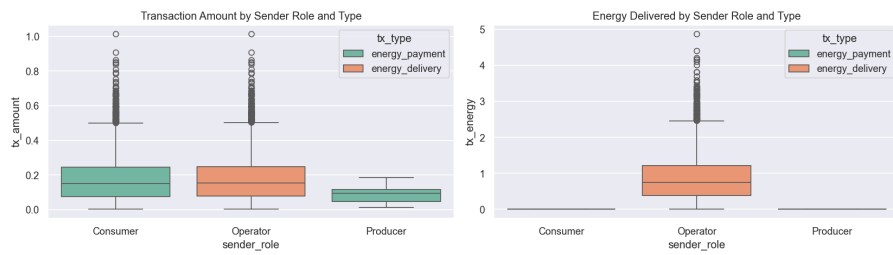


Figure 4.6: Transaction amount and energy distributions grouped by sender role and transaction type.

Across the 489 blocks, every block contained more than one transaction, averaging about 20.87 transactions per block (Figure 4.7). The spread of total payment volume across consumers (Figure 4.8) had a coefficient of variation of 0.586, indicating moderate but not extreme imbalance in trading activity.

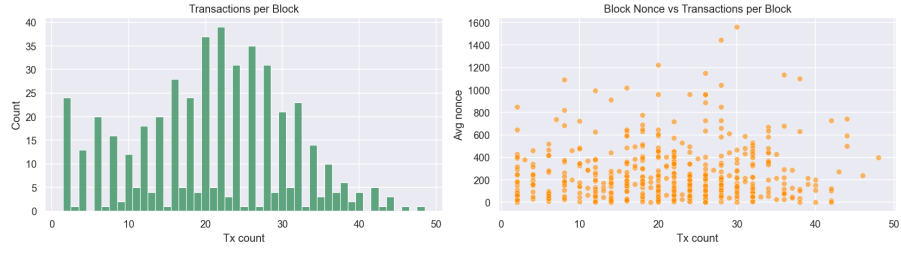


Figure 4.7: Distribution of transactions per block and relationship between block size and block nonce.

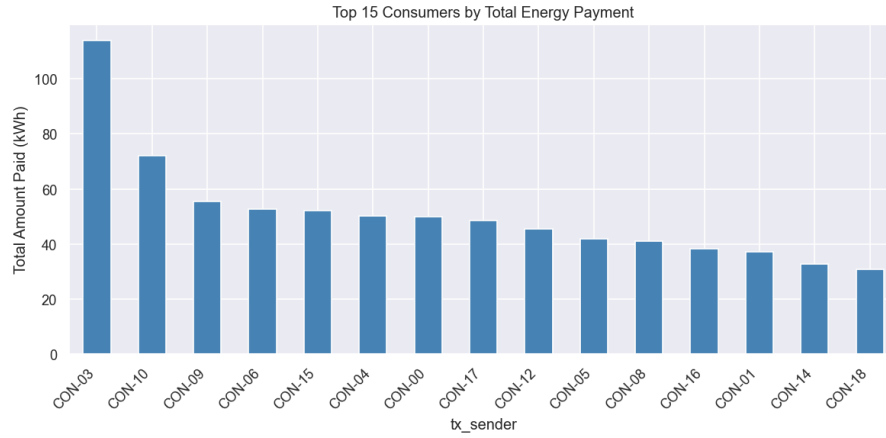


Figure 4.8: Top fifteen consumers ranked by total energy payment amount.

4.1.2 Feature Space and Anomaly Detection Results

Figure 4.9 shows the correlation matrix of the engineered features, and Figure 4.10 shows the two-dimensional PCA projection of the scaled feature space, which retained about 50.69% of the total variance.

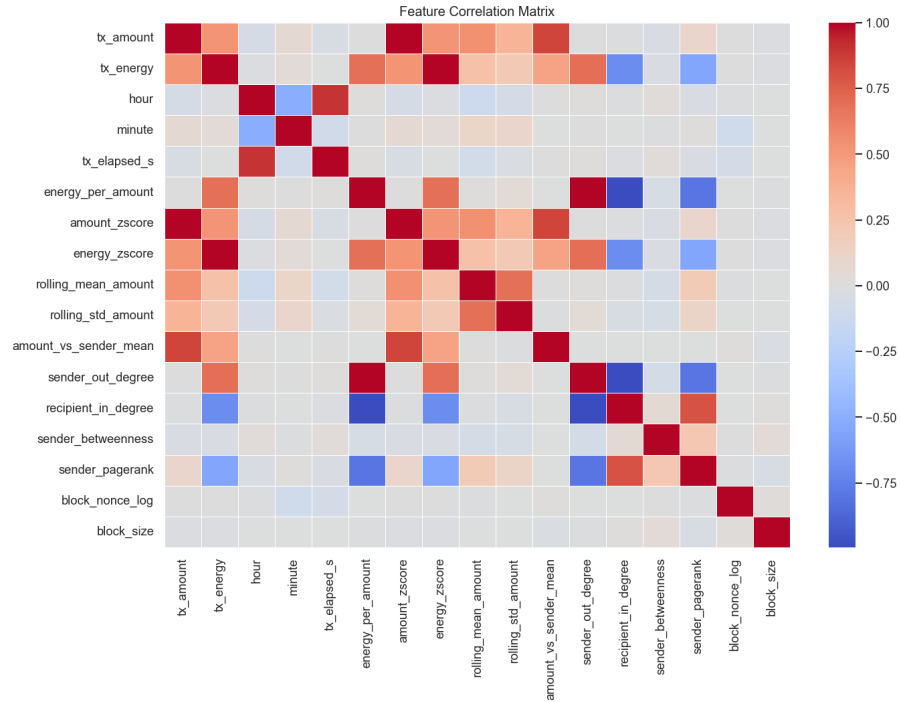


Figure 4.9: Correlation matrix of the engineered feature set.

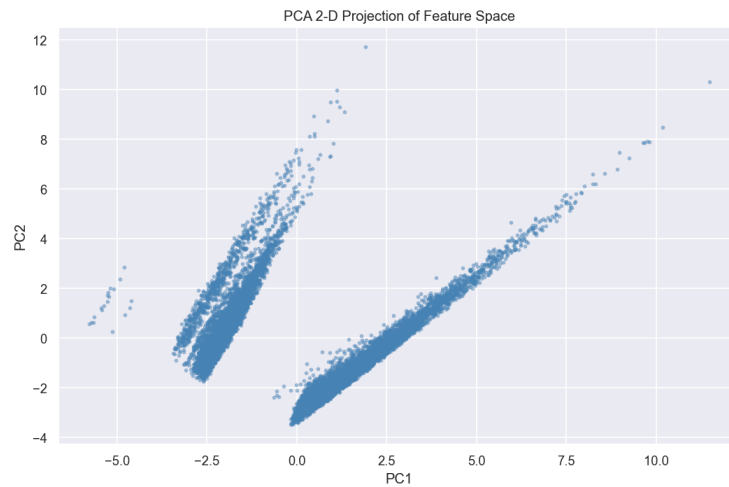


Figure 4.10: Two-dimensional PCA projection of the scaled feature space.

Out of 10,205 transactions, Isolation Forest flagged 154, LOF flagged 132, and One-Class SVM flagged 155 as anomalies. Applying the majority voting rule (flagged by at least two models), the ensemble identified 85 anomalous transactions: 48 energy_payment and 37 energy_delivery.

Figure 4.11 compares anomaly score distributions between normal and ensemble-flagged transactions. Across all three models, flagged transactions consistently re-

ceived lower (more anomalous) scores, showing agreement on what separates normal from anomalous behavior despite the models' different underlying methods.

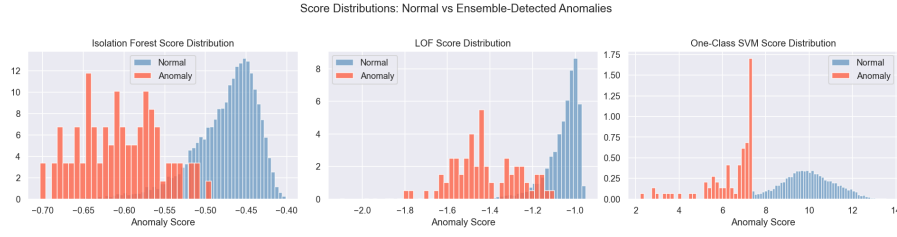


Figure 4.11: Distributions of anomaly scores from Isolation Forest, LOF, and One-Class SVM, compared between normal and ensemble-flagged anomalous transactions.

Figures 4.12 and 4.13 show the anomaly labels from each individual model and the final ensemble, respectively, in PCA space. Figure 4.14 shows the agreement matrix, measuring how consistently the three models flagged the same transactions.

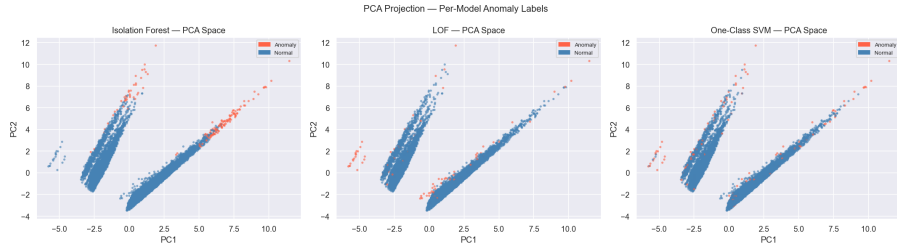


Figure 4.12: PCA projection showing anomaly labels assigned by each individual model.

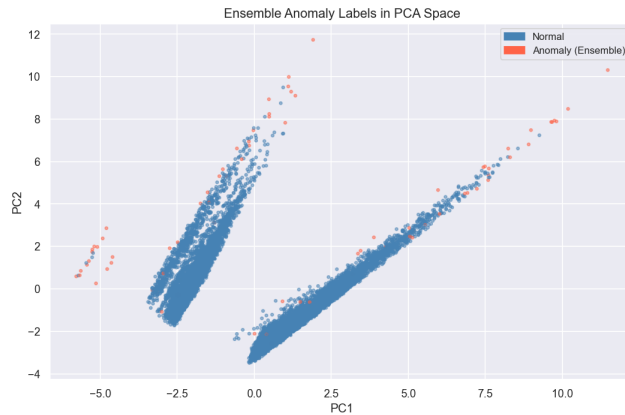


Figure 4.13: PCA projection showing the final ensemble anomaly labels.

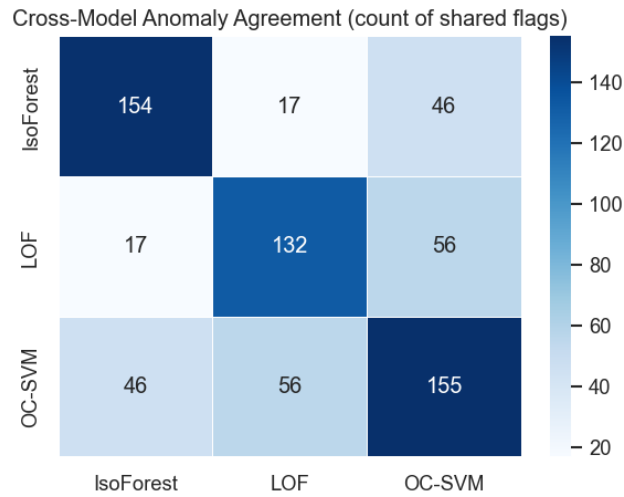


Figure 4.14: Agreement matrix showing the number of transactions flagged by each pair of models.

Figure 4.15 overlays the ensemble-flagged anomalies on the energy time series, showing their distribution over the observation period.

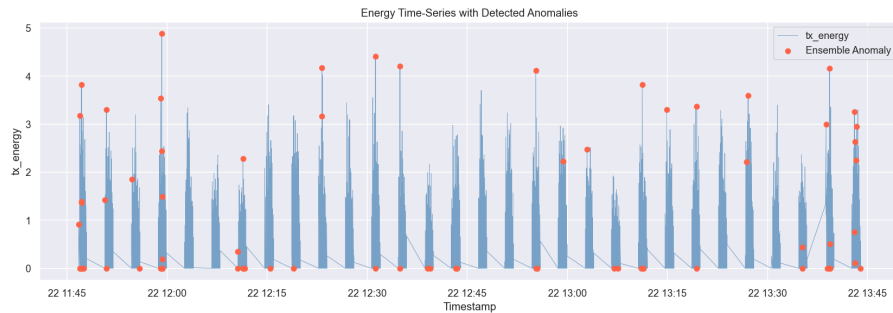


Figure 4.15: Ensemble-flagged anomalies overlaid on the energy time series.

4.2 Data Insights

The exploratory analysis shows a realistic P2P energy trading structure, with energy delivery transactions (carrying the actual energy value, sent only by the grid operator) clearly separated from energy payment transactions (carrying no energy value, representing settlement between consumers and producers). The hub-and-spoke structure, centered on the grid operator, makes this node critical, as failure or abnormal behavior here would likely propagate across the network.

The anomaly detection results support this. Among the 85 flagged transactions, GRID-OP-01 accounted for 37, the largest share, followed by CON-03 with 19 and PRO-07 with 11, showing anomalies concentrated around the central operator and a few other participants rather than spread evenly.

Comparing feature values between normal and anomalous transactions (Figure 4.16) shows `tx_energy` and `tx_amount` as the most distinguishing features, averaging about 2.50 and 2.35 times higher in anomalous transactions, while features like `energy_per_amount`, `block_nonce_log`, and `block_size` showed little difference (ratios close to 1), indicating the anomalies stem mainly from unusually large amounts and energy values rather than ratio or block-level irregularities.

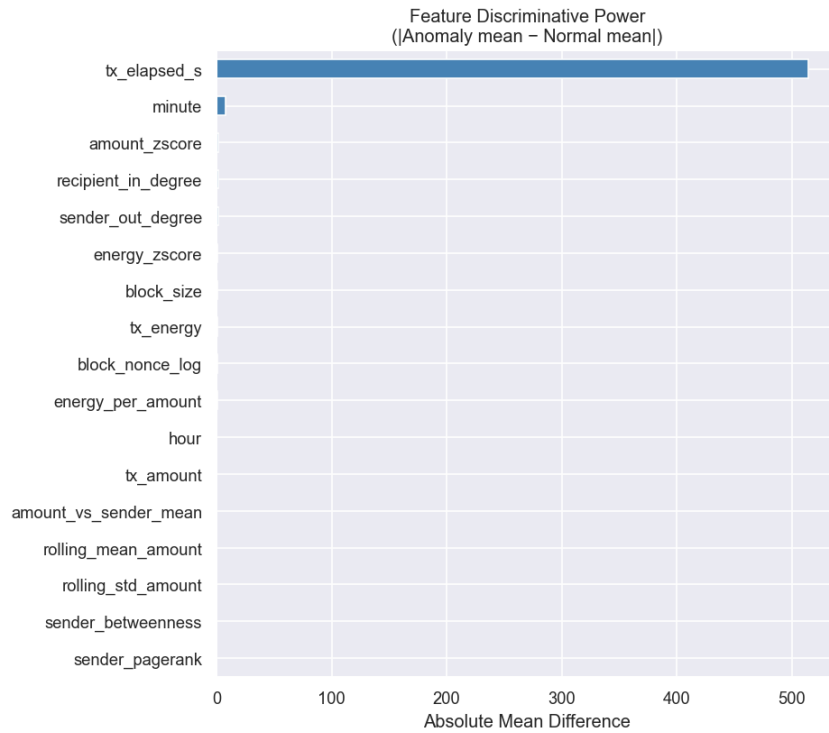


Figure 4.16: Absolute difference in mean feature values between normal and anomalous transactions, indicating feature discriminative power.

4.3 Model Interpretation

The three models approach anomaly detection differently, together giving a more complete picture than any one alone. Isolation Forest isolates transactions with relatively few splits, capturing extreme values across multiple features at once. Local Outlier Factor flags transactions in sparser regions than their neighbors, capturing local irregularities that may not appear extreme dataset-wide. One-Class SVM defines a boundary around the majority of the data using an RBF kernel and flags transactions outside it.

Despite their different principles, all three models consistently assigned lower anomaly scores to the same group of transactions (Figure 4.11), suggesting a real, consistent pattern rather than an artifact of any single algorithm. The ensemble voting strategy reinforces this, as each of the 85 final transactions was flagged by at

least two of the three models.

The concentration of anomalies around GRID-OP-01, combined with `tx_amount` and `tx_energy` as the most discriminative features, suggests the models mainly detect unusually large energy deliveries or transactions processed by the grid operator. In a real smart grid, such patterns could indicate sudden energy delivery spikes, an early sign of instability in power distribution, aligning with the project's goal of detecting deviations from normal trading behavior to flag potential operational issues.

Chapter V: Conclusion and Recommendations

5.1 Conclusion

This study applied an unsupervised ensemble framework, Isolation Forest, Local Outlier Factor, and One-Class SVM, to detect anomalous transactions in a simulated P2P smart grid blockchain dataset of 10,205 transactions across 489 blocks, with a transaction flagged as anomalous only when at least two models agreed.

The ensemble identified 85 anomalous transactions (about 0.83% of the dataset). Agreement across all three models, despite their different approaches, suggests these reflect genuine deviations rather than artifacts of a single algorithm. `tx_amount` and `tx_energy` were the most discriminative features, averaging about 2.35 and 2.50 times higher in anomalous transactions.

Anomalies were concentrated around the grid operator node (GRID-OP-01), which accounted for 37 of the 85 flagged transactions, pointing to unusual energy delivery spikes as the dominant anomaly pattern. Overall, combining diverse unsupervised detectors effectively surfaced suspicious transaction patterns even without labeled anomalies.

5.2 Recommendations

1. **Threshold calibration with expert review.** The 1.5% contamination rate should be validated through domain expert review of the flagged transactions against real-world anomaly rates.
2. **Transition to supervised learning.** The ensemble labels can serve as pseudo-labels for training supervised models such as XGBoost or LightGBM once verified.
3. **Real-time streaming deployment.** Isolation Forest could be deployed in streaming mode for real-time detection, with care to avoid temporal leakage in rolling statistics.
4. **Hash-chain integrity verification.** Rule-based checks on sequential `block_previous_hash` values would complement the statistical models by detecting block-level integrity violations.
5. **Validation on real-world datasets.** Testing the framework on real blockchain transaction data would strengthen generalizability and expose it to additional anomaly types.

References

- [1] Musleh, A. S., Yao, G., & Mueen, S. M. (2019). *Blockchain applications in smart grid: Review and frameworks*. IEEE Access, 7, 86746–86757.
- [2] Andoni, M., Robu, V., Flynn, D., Abram, S., Geach, D., Jenkins, D., McCallum, P., & Peacock, A. (2019). *Blockchain technology in the energy sector: A systematic review of challenges and opportunities*. Renewable and Sustainable Energy Reviews, 100, 143–174.
- [3] Li, Z., Kang, J., Yu, R., Ye, D., Deng, Q., & Zhang, Y. (2018). *Consortium blockchain for secure energy trading in industrial internet of things*. IEEE Transactions on Industrial Informatics, 14(8), 3690–3700.
- [4] Farrugia, S., Ellul, J., & Azzopardi, G. (2020). *Detection of illicit accounts over the Ethereum blockchain*. Expert Systems with Applications, 150, 113318.
- [5] Patel, V., et al. (2024). *Anomaly detection in blockchain: A systematic review of trends, challenges, and future directions*. Journal of Network and Computer Applications.
- [6] Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). *Isolation forest*. Proceedings of the IEEE International Conference on Data Mining (ICDM), 413–422.
- [7] Manevitz, L. M., & Yousef, M. (2001). *One-class SVMs for document classification*. Journal of Machine Learning Research, 2, 139–154.
- [8] Breunig, M. M., Kriegel, H. P., Ng, R. T., & Sander, J. (2000). *LOF: Identifying density-based local outliers*. Proceedings of the ACM SIGMOD International Conference on Management of Data, 93–104.
- [9] Campos, G. O., et al. (2016). *On the evaluation of unsupervised outlier detection: Measures, datasets, and an empirical study*. Data Mining and Knowledge Discovery, 30(4), 891–927.
- [10] Nguyen, D. C., et al. (2021). *Blockchain and AI: A review and open research challenges*. IEEE Access, 9, 110312–110332.

Appendix

Source Code

GitHub repository: github.com/RobbenWijanathan/fit-competition-2026

Dataset Details

Dataset link: [Track II Dataset Folder](#)